

Fault Code: 342

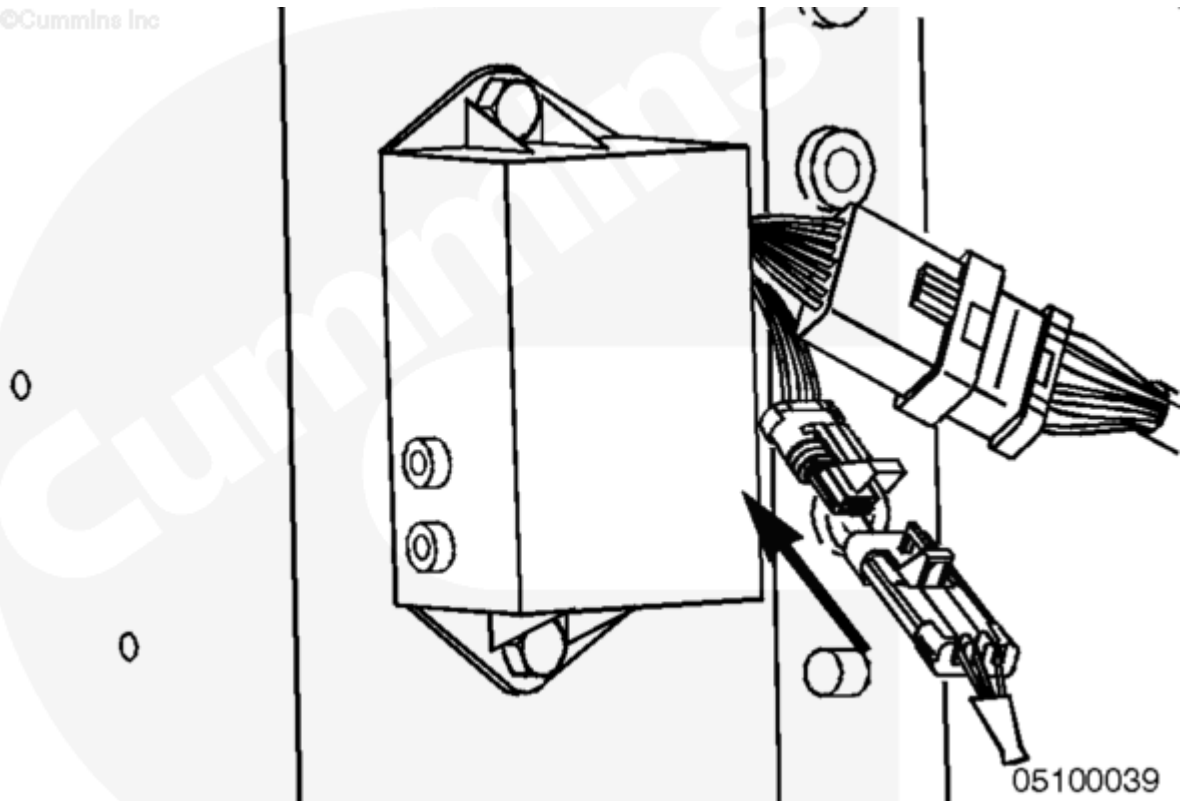
Centinel™ Control Module Calibration Plug Malfunctioning

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 342 PID(P): SPN: FMI: Lamp: SRT:	Centinel™ Control Module Calibration Plug Malfunctioning Heavy-Duty: no calibration plug installed. High-Horsepower: service plug was not removed.	The Centinel™ system will not operate properly.

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Circuit Description

The calibration plug is necessary for heavy-duty engines.

Component Location

- Heavy-Duty: The calibration plug is a 3-pin connector that is located on the Centinel™ control module.
- High-Horsepower: The service plug is a 3-pin connector in the wiring harness near the light box.

Warnings and Cautions

⚠ CAUTION ⚠

To avoid pin and harness damage, use the following test leads when taking measurements:
Male Cannon, Metri-Pack, and Deutsch test lead, Part Number 3822758
Female AMP, Metri-Pack, and Deutsch test lead, Part Number 3822917
Male Deutsch test lead, Part Number 3823993
Female Deutsch test lead, Part Number 3823994.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check calibration plug.	
	STEP 1A. Verify calibration plug is installed.	Install the calibration plug
	STEP 1B. Check for an open in the calibration plug.	Less than 10 ohms
STEP 2.	Check the Centinel™ harness.	
	STEP 2A. Check for an open circuit.	Less than 10 ohms
	STEP 2B. Check for a short circuit to ground.	More than 1k ohms
STEP 3.	Clear the fault codes.	
	STEP 3A. Disable the fault code.	Fault Code 342 inactive

STEP 1. Check the calibration plug.

STEP 1A. Verify calibration plug is installed.

Conditions :

- Turn the keyswitch OFF.
- Service plug removed.

Action	Specification/Repair	Next Step
• Inspect calibration plug.	Calibration plug is installed on heavy-duty. Service plug is not installed on high-horsepower.	1B
	Heavy-Duty: install the calibration plug High-Horsepower: remove the service plug.	3A

STEP 1B. Check for an open in the calibration plug.**Conditions :**

- Turn the keyswitch OFF.
- Remove the calibration plug from the Centinel™ control module.

Action	Specification/Repair	Next Step
• Heavy-Duty only: measure the resistance between pins that are connected. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Less than 10 ohms	2A
	Repair or replace the calibration plug	3A

STEP 2. Check the Centinel™ wiring harness.**STEP 2A. Check for an open circuit.**

Conditions :Heavy-Duty **only**:

- Turn the keyswitch OFF.
- Disconnect the Centinel™ harness from the Centinel™ control module.
- Remove the calibration plug from the Centinel™ control module.

Action	Specification/Repair	Next Step
• Measure the resistance from pin 7 of the 12-pin Centinel™ control module connector to pin B of the 3-pin calibration plug connector. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Less than 10 ohms	2B
	Replace the Centinel™ control module. Refer to Procedure 019-130 (/qs3/pubsys2/xml/en/procedures/96/96-019-130-tr.html).	3A

STEP 2B. Check for a short circuit to ground.**Conditions :**Heavy-duty **only**:

- Turn the keyswitch OFF.
- Remove the calibration plug from the Centinel™ harness.

Action	Specification/Repair	Next Step
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Measure the resistance from pin B of the 3-pin calibration plug (module side) to pins A and C. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 1k ohms	2D
	Replace the Sentinel™ wiring harness Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).	3A

STEP 3. Clear the fault codes.

STEP 3A. Disable the fault code.

Conditions :

- Connect all the components.

Action	Specification/Repair	Next Step
- Start the engine and let it idle for 1 minute. - Verify that Fault Code 342 is inactive.	Fault Code 342 inactive	Complete
	Return to troubleshooting steps or contact your local Cummins Authorized Repair Location if all the steps have been completed and checked again.	1A

